

Jegadeesan Karunakaran (Jag)

AI Architect | AI Strategy & Implementation Leader | Hands-on AI Engineer

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PROFESSIONAL SUMMARY

AI Architect and hands-on AI engineer with 25+ years of technology leadership across banking, government, aviation, healthcare, logistics, and real estate. Proven record of translating executive goals into Artificial Intelligence (AI) strategy and roadmaps, then delivering end to end: use-case prioritization, rapid prototyping, pilot-to-production rollout, and adoption tied to measurable KPIs and ROI. Deep expertise in LLM applications (Claude, GPT, Gemini), RAG, agentic automation, and AI governance, with production discipline in evaluation, observability, and cost control. Trusted by AIA, Citibank, Changi Airport Group, British Telecom, and Singapore government agencies.

CORE COMPETENCIES

AI Strategy & Governance: AI roadmaps · use-case portfolio prioritization · KPI & ROI baselines · governance gates · Responsible AI · executive stakeholder alignment · change management · digital transformation

GenAI & ML Engineering: LLM applications · RAG (Retrieval-Augmented Generation) · prompt engineering · embeddings & vector databases (pgvector) · LangChain · LangGraph · CrewAI · private LLMs (Ollama, llama.cpp) · applied machine learning

Agentic Automation: multi-agent systems · AI agents · workflow orchestration (n8n, Zapier, Make) · API integration · human-in-the-loop design · retries, error handling & monitoring

LLMOps & Observability: evaluation baselines & testing · logging & monitoring · cost and latency guardrails · explainability · change control

Engineering & Cloud: Python · TypeScript · JavaScript · FastAPI · Next.js · Node.js · .NET/C# · AWS · Azure · Google Cloud · Docker · Kubernetes · PostgreSQL · Redis · MongoDB

PROFESSIONAL EXPERIENCE

AI Architect · Hands-on AI Engineer | Singapore IT Services Pte Ltd

Singapore | 2023 – Present

- Partner with business leadership to define AI strategy and roadmaps; rank the AI use-case portfolio by impact, effort, and risk, and sequence pilots so early wins fund broader adoption.
- Designed and deployed multi-agent AI workflows (n8n, document parsers, custom LLM interfaces) automating real estate listing-to-close and back-office operations with human-in-the-loop validation, removing data-entry bottlenecks.
- Built structured LLM extraction pipelines converting unstructured emails, PDFs, and chat messages into validated JSON delivered to CRM and database APIs, with exception alerts and review checkpoints.
- Ship production GenAI applications in Python and TypeScript using Claude, GPT, and Gemini APIs, RAG pipelines, and vector databases; deploy private fine-tuned LLMs (Ollama, LoRA) where data must stay in-house.
- Instil production discipline: evaluation baselines, logging, monitoring, change control, and cost/latency guardrails keeping AI systems reliable, explainable, and auditable after go-live.
- Drive AI adoption with cross-functional teams via AI fluency workshops, prompt libraries, guardrails, and human checkpoints.

Digital Transformation Leader | Singapore IT Services Pte Ltd

Singapore | 2015 – 2022

- Led automation-first transformation consulting: mapped end-to-end business processes, then removed manual steps with automation, scripting, and cloud tooling before prescribing technology.
- Built machine learning and expert systems for Singapore real estate forecasting, plus autonomous systems covering the full marketing-to-completion transaction workflow.
- Defined outcomes, KPIs, and success metrics with stakeholders; scaled a specialist data-recovery operation 6x (12 to 74 cases per month) through process redesign and systemization.

Earlier Career – Software Engineering & Delivery Leadership | Singapore · USA

1999 – 2014

- Senior Systems Engineer, Singapore IT Services (2008–2014): designed secure on-premise and early cloud environments for enterprises across industries.
- Senior Lead Developer, Avanade Asia (2005–2008): led .NET enterprise delivery across APAC for British Telecom, AIA, MediaCorp, and Changi General Hospital; Top Employee Award. Earlier software roles in Singapore and the USA (1999–2005).

SELECTED ENTERPRISE CLIENTS

AIA · Citibank · Changi Airport Group · Ministry of Education (Singapore) · Public Service Division · British Telecom · Microsoft · Accenture · Nike · Lazard · Agility Logistics

CERTIFICATIONS

Anthropic Academy (2026): AI Fluency – Framework & Foundations · Claude Code · Claude Code in Action · Claude Cowork
WSQ Generative AI, SkillsFuture Singapore (2024) · Generative AI Fundamentals, Databricks (2024) · AI Productivity, Republic Polytechnic (2024)

Project Management Professional (PMP), PMI · Registered Management Consultant (RMC), Singapore · Software Engineering, IBM (2026)

EDUCATION

MBA, Information Technology – Annamalai University

2007

B.E., Electronics & Communication Engineering – National Engineering College, India · University Gold Medal

1999